

FISTRA: A Financial-Factor-Informed Set Transformer for Learning Tradable Factors from Stock Cross-Sections

Anonymous submission

Abstract

Deep factor models face a persistent trade-off between flexibility and interpretability: expressive latent representations can improve stock return prediction, but their factors often do not correspond to portfolios that can be formed and audited in practice. We propose FISTRA (A Financial-factor-Informed Set Transformer for learning **TR**adable factors), a conditional factor model that reconciles these goals by learning nonlinear cross-sectional structure while defining every latent factor as the return of an explicit stock portfolio. Rather than discarding established financial knowledge, FISTRA retains established financial factors in the return equation and learns complementary tradable factors beyond their span. It encodes each stock’s lagged history with temporal convolutions and uses induced set attention to model interactions across the stock cross-section. State-conditioned factor queries and a differentiable allocation layer then produce neutral, unit-gross portfolio weights before current returns are observed. The resulting factors are therefore replicable and auditable, while separate residual networks estimate characteristic-dependent exposures and strictly historical factor-return means generate out-of-sample predictions. Extensive experiments on real United States (U.S.) stock market data show that FISTRA outperforms state-of-the-art factor models in stock return prediction and investment performance. Further analyses demonstrate that its learned portfolios contain incremental information beyond established financial factors, achieving a favorable balance between predictive flexibility and operational interpretability.

Introduction

Factor models provide a compact account of common variation in stock returns and remain central to asset pricing and quantitative investment (Ross 1976; Fama and French 1993). Established factors—including market, size, value, profitability, investment, and momentum—are returns of portfolios with documented and reproducible construction rules (Fama and French 2015; Hou, Xue, and Zhang 2015; Jensen, Kelly, and Pedersen 2023). They encode decades of tested financial knowledge; ignoring them forces a learned model to rediscover known systematic structure from noisy returns before finding complementary signals.

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Machine learning permits characteristic-dependent exposures and expressive latent representations (Kelly, Pruitt, and Su 2019; Gu, Kelly, and Xiu 2021; Duan et al. 2022). However, a predictive latent coordinate need not correspond to any portfolio of real assets. We call a factor *tradable* when its stock weights are determined from information available before period t and its realized value is exactly $f_{k,t} = \sum_i w_{ik,t-1} r_{i,t}$. This operational definition guarantees ex-ante portfolio formation, exact replication, and direct auditing of holdings and stock-level trading costs even when the weight function is nonlinear.

Existing methods address only parts of this problem. Conditional autoencoder models produce portfolio factors but first compress the cross-section into a fixed set of characteristic-managed portfolios (Gu, Kelly, and Xiu 2021). Recent deep and prior-informed factor models learn richer representations, yet their latent variables do not directly specify portfolios formed before returns are observed (Duan, Wang, and Li 2025; Kim and Song 2026). Jointly retaining established financial knowledge, modeling the full stock cross-section, and learning directly tradable complementary factors therefore remains unresolved.

We propose FISTRA, a Financial-factor-Informed Set Transformer for learning **TR**adable factors directly from stock cross-sections. A conditional beta network estimates exposures to established and learned factors, while a temporal Set Transformer maps lagged stock histories to factor-specific portfolio scores. A differentiable long–neutral–short allocation converts these scores into stock weights, and exposure neutrality and temporal orthogonality discourage the learned factors from duplicating established ones. The weights are fixed before current returns enter, so every learned factor is exactly the realized return of a portfolio formed in advance. Figure 1 summarizes this information boundary.

FISTRA thus preserves validated financial structure without assuming that established factors span all systematic returns. Its learned factors remain flexible but can be reconstructed from saved stock weights, while lagged factor-return estimates support prediction without future information.

Our contributions are fourfold:

- We formulate a financial-factor-informed conditional factor model that preserves established knowledge while directing learning toward complementary systematic struc-

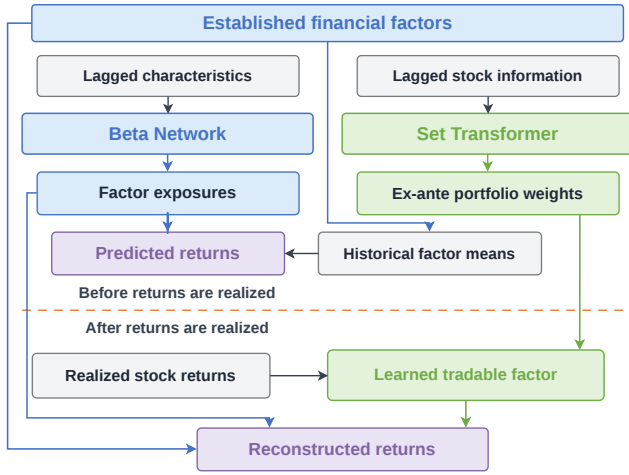


Figure 1: Brief illustration of FISTRA.

ture.

- We develop a temporal Set Transformer that captures stock dynamics and permutation-equivariant cross-stock interactions without managed-portfolio compression.
- We embed structured allocation and established-factor neutrality in learning, making every factor directly reproducible from positions fixed before returns and reducing black-box ambiguity.
- We conduct extensive experiments on U.S. stocks. FISTRA outperforms state-of-the-art factor models in prediction and investment simulation; ablation, spanning, and robustness tests support its main design choices.

Related Work

Financial and Conditional Factor Models

Classical models define factors through transparent portfolio rules, while large-scale replications organize published anomalies into established themes (Fama and French 1993, 2015; Hou, Xue, and Zhang 2015; Jensen, Kelly, and Pedersen 2023). Latent models instead estimate the factor space: principal component analysis (PCA) extracts covariance directions, instrumented principal component analysis (IPCA) uses characteristics to model time-varying loadings, and Autoencoder Asset Pricing (Autoencoder AP) learns nonlinear loadings from characteristic-managed portfolios (Bai and Ng 2002; Kelly, Pruitt, and Su 2019; Gu, Kelly, and Xiu 2021). Evidence that omitted factors affect inference and that established and latent spaces need not coincide motivates modeling both jointly (Giglio and Xiu 2021; Andreou et al. 2025). FISTRA follows this conditional factor perspective but learns additional portfolios at the stock level rather than from a compressed managed-portfolio panel.

Deep Latent Factor Models for Stock Returns

Deep factor models use variational, graph, contrastive, adversarial, and vector-quantized representations (Duan et al. 2022; Wei et al. 2023; Jia et al. 2024; Duan, Wang, and Li 2025; Liu, Huang, and Ruf 2025; Kim, Ock, and Song 2025).

NeurIF adds characteristic-controlled spatial-temporal attention, but forms long-short portfolios only after learning latent factors and loadings; STORM’s multidimensional discrete embeddings likewise require downstream trading construction (Uddin 2025; Zhao et al. 2026). Feng et al. (2024) learn a shared stockwise nonlinear transformation and then apply a fixed long-short sorting rule, without contextual cross-stock interaction or explicit established-beta neutrality. FactorGCL uses coarse stock-prototype loadings such as industries rather than established factor returns, while PRISM-VQ conditions stock codes on expert factor histories; neither outputs factor portfolios (Duan, Wang, and Li 2025; Kim and Song 2026). FISTRA instead learns contextual stock weights fixed in advance.

Set Learning for Stock Cross-Sections

Stock Transformers such as DTML, MASTER, and CISTHPAN model temporal and cross-stock dependence but output forecasts or rankings rather than factor returns and portfolios (Yoo et al. 2021; Li et al. 2024; Xia et al. 2024). Deep Sets formalizes permutation-invariant functions, while Set Transformer models interactions efficiently through induced attention (Zaheer et al. 2017; Lee et al. 2019). This suits variable, unordered stock cross-sections; FISTRA returns context to every stock to generate factor-specific scores.

Preliminaries

Established Financial Factors

An established financial factor is a portfolio return generated by a documented rule. The Fama–French three-factor model (FF3), for example, contains market, size, and value factors; size compares small with large firms, while value compares high with low book-to-market firms (Fama and French 1993). The 13 themes of Jensen, Kelly, and Pedersen (2023) provide a broader summary of published return predictors.

Let $P_{i,t}$ be the end-of-period price of stock i and $D_{i,t}$ be its cash distribution during period t . The total holding-period return is

$$r_{i,t} = \frac{P_{i,t} + D_{i,t} - P_{i,t-1}}{P_{i,t-1}}. \quad (1)$$

Our target is the delisting-adjusted return, including cash distributions. For month t , let $\mathcal{U}_t$ be the stock universe eligible at the beginning of the month, $N_t = |\mathcal{U}_t|$, and

$$r_t = (r_{1,t}, \dots, r_{N_t,t})' \in \mathbb{R}^{N_t}. \quad (2)$$

Let K^o denote the number of established factors and let $g_t \in \mathbb{R}^{K^o}$ collect their returns. Each component of g_t is itself generated by portfolio positions fixed before month t ; the vector is common to all stocks in that month.

Problem Formulation

Let $z_{i,t-1} \in \mathbb{R}^P$ contain firm characteristics available before month t , and stack the eligible stocks rowwise as $Z_{t-1} = [z'_{i,t-1}]_{i \in \mathcal{U}_t} \in \mathbb{R}^{N_t \times P}$. We model returns as

$$r_{i,t} = b^o(z_{i,t-1}, \theta_o)' g_t + b^h(z_{i,t-1}, \theta_h)' f_t + \varepsilon_{i,t}, \quad (3)$$

Here K^h denotes the number of learned factors, and superscripts o and h index the observed established and learned

hidden branches, respectively. The vectors $b^o \in \mathbb{R}^{K^o}$ and $b^h \in \mathbb{R}^{K^h}$ are the corresponding conditional exposures, θ_o and θ_h are their network parameters, and $\varepsilon_{i,t}$ is the residual. For $a \in \{o, h\}$, $B^a(Z_{t-1}, \theta_a)$ denotes the matrix that stacks $b^a(z_{i,t-1}, \theta_a)'$ over $i \in \mathcal{U}_t$.

For every stock, define a length- L information history

$$H_{i,t-1} = \{[z_{i,s-1}, r_{i,s-1}] : s = t - L + 1, \dots, t\}. \quad (4)$$

The complete information set is the variable-size stock set

$$I_{t-1} = \{H_{i,t-1} : i \in \mathcal{U}_t\}. \quad (5)$$

FISTRA learns a portfolio-weight function satisfying

$$W(I_{t-1}, \phi) \in \mathbb{R}^{N_t \times K^h}, \quad f_t = W(I_{t-1}, \phi)' r_t. \quad (6)$$

Here ϕ denotes the factor-network parameters. Equation (6) is the key information and tradability requirement: $W(I_{t-1}, \phi)$ is measurable before period- t returns, and r_t enters only to settle the realized returns of already specified portfolios.

Methodology

Model Overview

Figure 2 summarizes the complete architecture. FISTRA contains a conditional beta network, a temporal Set Transformer, a structured tradable factor layer, and a historical-mean estimator for expected factor returns. Two separate beta networks map Z_{t-1} to $B^o(Z_{t-1}, \theta_o) \in \mathbb{R}^{N_t \times K^o}$ and $B^h(Z_{t-1}, \theta_h) \in \mathbb{R}^{N_t \times K^h}$. The temporal Set Transformer maps I_{t-1} to stock-factor scores through stockwise temporal encoding, induced set attention, and state-conditioned factor queries. The structured layer converts these scores into $W(I_{t-1}, \phi)$ under explicit portfolio constraints. Applying the resulting weights to realized stock returns yields f_t . The contemporaneous fitted return is

$$\hat{r}_t^{\text{fit}} = B^o(Z_{t-1}, \theta_o) g_t + B^h(Z_{t-1}, \theta_h) f_t. \quad (7)$$

For prediction, strictly lagged estimates of expected established and learned factor returns replace their current realizations.

Conditional Beta Network

Established and hidden factor exposures have different economic roles, so FISTRA uses two residual multilayer perceptrons (MLPs) with the same form but unshared parameters. For branch $a \in \{o, h\}$, an input projection produces $x_{i,a}^{(0)}$ from $z_{i,t-1}$. Using layer normalization (LN), residual block ℓ computes

$$x_{i,a}^{(\ell+1)} = x_{i,a}^{(\ell)} + \mathcal{D}\left(M_{2,a}^{(\ell)} \sigma(M_{1,a}^{(\ell)} \text{LN}(x_{i,a}^{(\ell)}))\right), \quad (8)$$

where σ is a nonlinear activation and $\mathcal{D}$ is dropout. A final normalized linear projection outputs $b^o(z_{i,t-1}, \theta_o)$ or $b^h(z_{i,t-1}, \theta_h)$. Residual connections stabilize optimization while preserving a direct path from characteristics to factor exposures (He et al. 2016).

Temporal Set Transformer

Causal Temporal Encoding FISTRA first encodes each $H_{i,t-1}$ independently with a temporal convolutional network (TCN) comprising left-padded dilated convolutions (Bai, Kolter, and Koltun 2018):

$$u_{i,t-1} = \text{TCN}(H_{i,t-1}) \in \mathbb{R}^d. \quad (9)$$

Here d is the representation dimension. At every intermediate position, left-only padding prevents access to later observations. Retaining the final temporal state summarizes the stock's lagged characteristics and return dynamics.

Cross-Sectional Set Interaction An induced set attention block (ISAB) with m inducing queries is defined using multihead attention blocks (MABs) for inner aggregation and outer broadcast. Let $U_{t-1}^{(0)} = U_{t-1} = \{u_{i,t-1}\}_{i=1}^{N_t}$; the two stacked blocks compute

$$\begin{aligned} \text{ISAB}_m(X, Q_m) &= \text{MAB}(X, \text{MAB}(Q_m, X)), \\ U_{t-1}^{(\ell)} &= \text{ISAB}_m(U_{t-1}^{(\ell-1)}, Q_m^{(\ell)}), \quad \ell = 1, 2, \end{aligned} \quad (10)$$

and set $\tilde{U}_{t-1} = U_{t-1}^{(2)}$. Here $Q_m^{(\ell)} \in \mathbb{R}^{m \times d}$ contains m learned inducing queries. Each MAB uses multihead attention (MHA) and a shared rowwise feed-forward network (rFF):

$$\begin{aligned} A(X, Y) &= \text{LN}(X + \text{MHA}(X, Y, Y)), \\ \text{MAB}(X, Y) &= \text{LN}(A(X, Y) + \text{rFF}(A(X, Y))), \end{aligned} \quad (11)$$

where X supplies the queries and Y supplies the keys and values (Lee et al. 2019). Multihead attention, residual connections, layer normalization, and the rFF constitute the Transformer operation. The first MAB compresses the stock set into m summaries, and the second broadcasts that context back to every stock. With no stock-order positional encoding, the map is permutation equivariant and has principal attention cost $O(N_t m)$ rather than $O(N_t^2)$. Thus the preceding causal TCN supplies temporal encoding, whereas the ISAB layers are the Set Transformer operating across the same-month stock set.

State-Conditioned Factor Queries Each learned factor $k \in \{1, \dots, K^h\}$ has a base query $q_k^0 \in \mathbb{R}^d$. To allow portfolio composition to vary with the cross-sectional state, FISTRA adjusts the query using the mean contextual representation available at $t-1$:

$$\bar{u}_{t-1} = \frac{1}{N_t} \sum_i \tilde{u}_{i,t-1}, \quad q_{k,t-1} = q_k^0 + h_k(\bar{u}_{t-1}), \quad (12)$$

$$s_{ik,t-1} = \frac{\kappa(\tilde{u}_{i,t-1})' q_{k,t-1}}{\sqrt{d}}. \quad (13)$$

Here h_k is a state-adjustment map and $\kappa(\cdot)$ is a learned key projection. The score $s_{ik,t-1}$ is relative to the eligible stock set for period t but uses only information available at $t-1$.

Structured Tradable Factor Layer

Established-Factor Neutrality To discourage the learned portfolios from duplicating the established branch, let

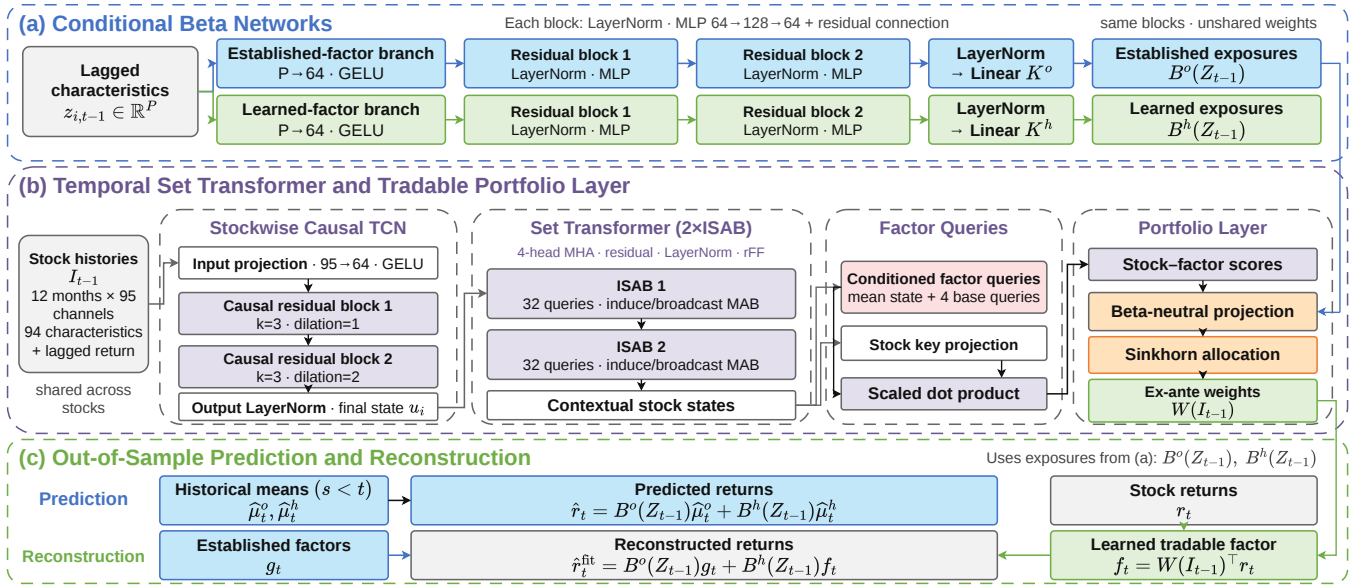


Figure 2: Architecture of FISTRA. (a) Conditional beta networks estimate exposures to established and learned factors. (b) The temporal Set Transformer maps lagged stock histories to ex-ante tradable portfolio weights. (c) Historical factor-return estimates produce out-of-sample predictions, while realized factor returns reconstruct contemporaneous stock returns.

$\tilde{B}^o(Z_{t-1}, \theta_o)$ denote the column-standardized version of $B^o(Z_{t-1}, \theta_o)$ and define

$$C_{t-1} = [\mathbf{1}, \tilde{B}^o(Z_{t-1}, \theta_o)] \in \mathbb{R}^{N_t \times (K^o + 1)}. \quad (14)$$

For the score matrix $S_{t-1} = (s_{ik,t-1})$, FISTRA applies a regularized cross-sectional projection

$$S_{t-1}^\perp = S_{t-1} - C_{t-1}(C_{t-1}'C_{t-1} + \lambda_n I_{K^o+1})^{-1}C_{t-1}'S_{t-1}. \quad (15)$$

where I_{K^o+1} is the identity matrix and $\lambda_n > 0$ is a fixed numerical stabilizer for the Gram-matrix solve, rather than a tuned prediction penalty; its value appears in Implementation Details. The beta matrix is detached on this path, preventing the loading network from changing its established-factor exposures merely to weaken the portfolio constraint. The projection produces approximately zero net scores and approximately zero exposure to the estimated established-factor betas.

Differentiable Long-Short Allocation FISTRA maps each projected score to differentiable soft assignments over short, neutral, and long states. The Sinkhorn operator iteratively rescales a positive matrix to prescribed row and column marginals (Cuturi 2013); we use equal short and long budgets to encourage self-financing portfolios and a neutral state to allow inactive positions. A stock's signed weight is its long mass minus its short mass. After a final exposure-neutral projection, each factor is normalized to unit gross exposure. The supplementary appendix gives the logits, marginals, and normalization in full.

Factor Realization Only after $W(I_{t-1}, \phi)$ is fixed do current returns enter the model:

$$f_{k,t} = \sum_{i \in \mathcal{U}_t} w_{ik}(I_{t-1}, \phi) r_{i,t}. \quad (16)$$

Thus each hidden factor is not merely correlated with a portfolio return; it is exactly that return. Saving the stock-level weights makes the factor independently reproducible and exposes its holdings, concentration, and transaction-cost implications. This construction preserves a nonlinear mapping from information to weights while avoiding the black-box problem of latent coordinates that cannot be directly traded.

Objective Function

The main training objective minimizes contemporaneous cross-sectional reconstruction error:

$$\mathcal{L}_{\text{fit}} = \sum_t \frac{1}{N_t} \|r_t - B^o(Z_{t-1}, \theta_o)g_t - B^h(Z_{t-1}, \theta_h)f_t\|_2^2. \quad (17)$$

To prevent different queries from producing nearly identical portfolios, let $\bar{w}_{k,t-1} = w_k(I_{t-1}, \phi) / \|w_k(I_{t-1}, \phi)\|_2$. For $K^h > 1$, define

$$\mathcal{L}_{\text{div}} = \sum_t \frac{\sum_{k \neq l} (\bar{w}'_{k,t-1} \bar{w}_{l,t-1})^2}{K^h(K^h - 1)}. \quad (18)$$

We set $\mathcal{L}_{\text{div}} = 0$ when $K^h = 1$. Cross-sectional beta neutrality is period-specific; we additionally discourage persistent time-series replication of established factors. Let $\mathcal{T}_{\text{orth}}$ contain the selected training months and $T_{\text{orth}} = |\mathcal{T}_{\text{orth}}|$. For standardized factor matrices $\tilde{F} \in \mathbb{R}^{T_{\text{orth}} \times K^h}$ and $\tilde{G} \in \mathbb{R}^{T_{\text{orth}} \times K^o}$, we use

$$\mathcal{L}_{\text{orth}} = \frac{1}{K^h K^o} \left\| \frac{1}{T_{\text{orth}}} \tilde{F}' \tilde{G} \right\|_F^2. \quad (19)$$

The conceptual objective is

$$\mathcal{L} = \mathcal{L}_{\text{fit}} + \lambda_{\text{div}} \mathcal{L}_{\text{div}} + \lambda_{\text{orth}} \mathcal{L}_{\text{orth}}. \quad (20)$$

where $\lambda_{\text{div}}, \lambda_{\text{orth}} \geq 0$ control portfolio diversity and time-series orthogonality.

Out-of-Sample Prediction

Contemporaneous factor realizations in Equation (7) are used for factor estimation, not for predicting their own month. For test month t , FISTRA estimates expected factor returns only from earlier realizations:

$$\hat{\mu}_t^o = \frac{1}{|\mathcal{T}_t|} \sum_{s \in \mathcal{T}_t} g_s, \quad \hat{\mu}_t^h = \frac{1}{|\mathcal{T}_t|} \sum_{s \in \mathcal{T}_t} f_s, \quad (21)$$

where $\mathcal{T}_t = \{s : s < t\}$ contains all factor returns realized and available by the prediction date. The annually estimated model is fixed throughout each test year, while returns from earlier test months enter these historical means sequentially once realized. Thus $\hat{\mu}_t^o$ and $\hat{\mu}_t^h$ estimate expected established and learned factor returns without month- t information. The one-month-ahead prediction is

$$\hat{r}_{i,t} = b^o(z_{i,t-1}, \theta_o)' \hat{\mu}_t^o + b^h(z_{i,t-1}, \theta_h)' \hat{\mu}_t^h. \quad (22)$$

Accordingly, $\hat{r}_{i,t}$ estimates stock i 's expected return in month t . Following Autoencoder AP (Gu, Kelly, and Xiu 2021), applicable factor models share the same parameter-free expanding mean, deliberately holding forecasting complexity fixed so differences primarily reflect factor and exposure learning.

Experiments

Experimental Setup

Research Questions Our experiments address four research questions (RQs):

- *RQ1 (Overall effectiveness)*. Does FISTRA outperform strong baselines in both cross-sectional return prediction and downstream investment performance?
- *RQ2 (Component validity)*. How do established financial factors, cross-stock set interaction, state-conditioned factor queries, and structured long-short allocation contribute to performance?
- *RQ3 (Incremental information)*. Do the learned factor portfolios contain realized-return information beyond the established financial factor span?
- *RQ4 (Robustness)*. Are the conclusions stable across model capacities and transaction-cost assumptions?

Data and Protocol We use an unbalanced panel of U.S. common stocks from January 1980 to December 2025. The target is the monthly Center for Research in Security Prices (CRSP) return with delisting adjustments and cash distributions; predictors are 94 point-in-time firm characteristics from Gu, Kelly, and Xiu (2020), cross-sectionally rank normalized to $[-1, 1]$. Stock eligibility is determined solely by information available at the beginning of each month; if an eligible stock exits during that month, its delisting-adjusted return remains in the sample. Established-factor inputs are either the 13 value-weighted U.S. themes of Jensen, Kelly, and Pedersen (2023) (JKP13) or the market, size, and value factors of Fama and French (1993) (FF3), with FF3 constructed on the same stock universe using lagged market equity.

From January 2010 to December 2025, we use annual expanding windows: before each test year, the preceding ten years form the validation set, all observations before that period form the training set, and the selected model is fixed for the entire test year. All preprocessing, estimation, and portfolio formation respect the same pre-test information cutoff.

Comparison Methods To reflect our factor-model setting, we compare representative classical factor baselines—FF3, JKP13, PCA, and IPCA (Fama and French 1993; Jensen, Kelly, and Pedersen 2023; Bai and Ng 2002; Kelly, Pruitt, and Su 2019)—and state-of-the-art neural factor models—Autoencoder AP, FactorVAE, FactorGCL, and CAVB (Gu, Kelly, and Xiu 2021; Duan et al. 2022; Duan, Wang, and Li 2025; Liu, Huang, and Ruf 2025). We also include the recent prior-informed non-factor ranking model PRISM-VQ (Kim and Song 2026). FISTRA-FF3 and FISTRA-JKP13 use the compact FF3 and broad JKP13 factor sets, respectively. All methods share the same protocol, and applicable latent-factor models use $K^h = 4$.

Evaluation Metrics Monthly information coefficient (IC) and rank information coefficient (RankIC) are the cross-sectional Pearson and Spearman correlations between predicted and realized returns; we report their time-series means and mean-to-standard-deviation ratios (ICIR and RankICIR). Here $\hat{r}_{i,t}$ is the prediction in Equation (22), and $\hat{r}_{i,t}^{\text{fit}}$ is the fitted return in Equation (7). Following Gu, Kelly, and Xiu (2021), we compute

$$R_{\text{pred}}^2 = 1 - \frac{\sum_{i,t} (r_{i,t} - \hat{r}_{i,t})^2}{\sum_{i,t} r_{i,t}^2}, \quad (23)$$

$$R_{\text{total}}^2 = 1 - \frac{\sum_{i,t} (r_{i,t} - \hat{r}_{i,t}^{\text{fit}})^2}{\sum_{i,t} r_{i,t}^2}.$$

R_{pred}^2 measures forecasting accuracy, whereas R_{total}^2 measures contemporaneous explanatory power. Investment metrics are annualized return (AR), cumulative return (CR), Sharpe ratio (SR), and maximum drawdown (MDD).

Implementation Details Each stock input contains a 12-month history of 94 characteristics and one strictly lagged return channel. The beta network uses separate residual multilayer perceptron (MLP) branches, while the factor network combines causal temporal convolutions, two ISAB layers, and four state-conditioned queries. We fix $\lambda_n = 10^{-4}$ and $\lambda_{\text{div}} = \lambda_{\text{orth}} = 0.01$ across vintages and priors. Models are optimized with Adam and selected by validation MSE; we report results averaged over five random seeds. Complete architectural, optimization, and reproducibility settings are provided in the supplementary appendix.

Overall Performance (RQ1)

Table 1 summarizes six predictive metrics over the common 192-month test sample and shows that FISTRA leads every metric. FISTRA-JKP13 attains the highest IC (3.709%), ICIR (0.543), R_{pred}^2 (1.340%), and R_{total}^2 (10.406%), improving over the strongest non-FISTRA result by 29.0%, 27.2%, 76.3%, and 4.0%, respectively. This advantage indicates that the broad JKP13 prior supports accurate and stable return-level prediction and reconstruction. FISTRA-FF3

Model	IC (%)	ICIR	RankIC (%)	RankICIR	R^2_{pred} (%)	R^2_{total} (%)
FF3	0.292	0.042	1.354	0.130	-0.574	3.288
JKP13	0.026	0.008	0.570	0.132	-3.028	4.374
PCA	0.319	0.081	0.565	0.056	-5.756	6.267
IPCA	2.172	0.368	1.836	0.344	0.620	8.883
Autoencoder AP	2.876	0.427	2.652	0.379	0.760	10.008
FactorVAE	2.009	0.335	2.706	0.314	0.320	8.042
FactorGCL	1.795	0.225	2.710	0.326	-0.482	–
CAVB	2.819	0.258	4.295	0.445	0.520	5.187
PRISM-VQ	1.895	0.181	2.151	0.319	0.152	–
FISTRA-FF3	3.324	0.491	5.894	0.554	1.030	9.887
FISTRA-JKP13	3.709	0.543	5.295	0.511	1.340	10.406

Table 1: Overall Out-of-Sample Predictive Performance

achieves the best RankIC (5.894%) and RankICIR (0.554), exceeding CAVB by 37.2% and 24.5%; thus, even the compact FF3 prior enables more accurate and stable cross-sectional ranking. Together, the two specifications show that FISTRA’s gains persist across established-factor sets while their information breadth affects which prediction objective benefits most. Dashes denote unavailable R^2_{total} : FactorGCL and PRISM-VQ provide return predictions but do not define a comparable contemporaneous reconstructed return.

Investment Simulation (RQ1)

Stock-Selection Portfolios Each month, we form equal-weighted portfolios from predicted-return deciles: long the top and short the bottom, or long only the top. Table 2 reports AR, CR, SR, and MDD net of 25-bp trading costs; the S&P 500 is the total-return benchmark.

FISTRA-JKP13 achieves the best model-based AR in both long–short (14.86%, SR 1.15) and long-only (21.84%, CR 1,830.75%) portfolios, and Figure 3 shows the highest final wealth in both panels. These results show that FISTRA’s predictive gains survive portfolio formation and 25-bp transaction costs, translating statistical accuracy into economically meaningful out-of-sample investment performance. The consistent advantage under both long–short and long-only rules further indicates that its stock ranking remains useful under different investment constraints, while its leading long–short SR demonstrates the strongest risk-adjusted performance among the compared models.

Tradable Factor Portfolio Allocation Because each FISTRA factor is a portfolio formed before returns are realized, it can enter investment allocation directly rather than first being converted from stock predictions into a top-minus-bottom strategy. At each test-year start, we construct matched mean–variance portfolios using FF3 or JKP13 alone and after adding four FISTRA factors. Factor scaling, historical means, and Ledoit–Wolf covariances use only the expanding pre-test history (Markowitz 1952; Ledoit and Wolf 2004); allocations have a 10% ex ante volatility ceiling and total absolute factor weights at most two. Factor allocations are fixed for the test year, while FISTRA’s stock weights are recomputed monthly from lagged information. We aggregate the learned-factor sleeves into stock positions and deduct 25-bp trading costs

Panel A: Long-short				
Method	AR (%)	CR (%)	Sharpe	MDD (%)
FF3	3.62	70.44	0.48	-15.15
JKP13	4.07	85.92	0.66	-9.27
PCA	2.78	49.13	0.37	-16.61
IPCA	8.60	252.24	0.71	-26.47
Autoencoder AP	12.86	569.06	0.93	-18.20
FactorVAE	12.01	462.68	0.77	-31.30
FactorGCL	5.01	107.58	0.53	-35.54
CAVB	10.80	282.30	0.51	-43.99
PRISM-VQ	2.46	35.27	0.23	-22.11
FISTRA-FF3	12.23	519.50	0.96	-15.39
FISTRA-JKP13	14.86	836.34	1.15	-15.86

Panel B: Long-only				
Method	AR (%)	CR (%)	Sharpe	MDD (%)
FF3	14.20	598.47	0.71	-29.09
JKP13	14.84	687.44	0.76	-30.19
PCA	15.06	686.38	0.73	-32.19
IPCA	17.05	981.76	0.83	-28.89
Autoencoder AP	20.87	1569.78	0.82	-37.62
FactorVAE	19.12	1169.15	0.75	-34.75
FactorGCL	16.55	942.30	0.87	-35.37
CAVB	17.09	932.23	0.76	-29.55
PRISM-VQ	15.26	733.15	0.77	-35.92
FISTRA-FF3	20.27	1452.12	0.81	-33.73
FISTRA-JKP13	21.84	1830.75	0.85	-34.23
S&P 500	14.31	728.32	1.00	-23.87

Table 2: Stock-selection performance net of 25 bps, 2010–2025.

Panel A: FF3-Based Factor Portfolios				
Method	AR (%)	CR (%)	Sharpe	MDD (%)
FF3	7.81	215.29	0.71	-25.34
FF3 + FISTRA-FF3	16.42	1166.28	1.73	-11.88

Panel B: JKP13-Based Factor Portfolios				
Method	AR (%)	CR (%)	Sharpe	MDD (%)
JKP13	4.59	98.31	0.59	-21.43
JKP13 + FISTRA-JKP13	10.58	403.46	1.13	-22.40

Table 3: Tradable factor-portfolio performance; augmented returns are net of 25-bp learned-factor stock-rebalancing costs.

from their monthly rebalancing; the supplementary appendix gives the exact calculation.

Table 3 shows that the learned factors remain useful after charging costs to their actual stock holdings. Adding FISTRA-FF3 raises AR from 7.81% to 16.42%, SR from 0.71 to 1.73, and roughly halves the magnitude of MDD from 25.34% to 11.88%. FISTRA-JKP13 likewise more than doubles AR and nearly doubles SR. Unlike stock selection, this experiment evaluates the learned factors themselves: their explicit holdings permit costs to be deducted directly, and the resulting returns, Sharpe ratios, and drawdown control demonstrate a practical advantage of tradable rather than purely latent factors.

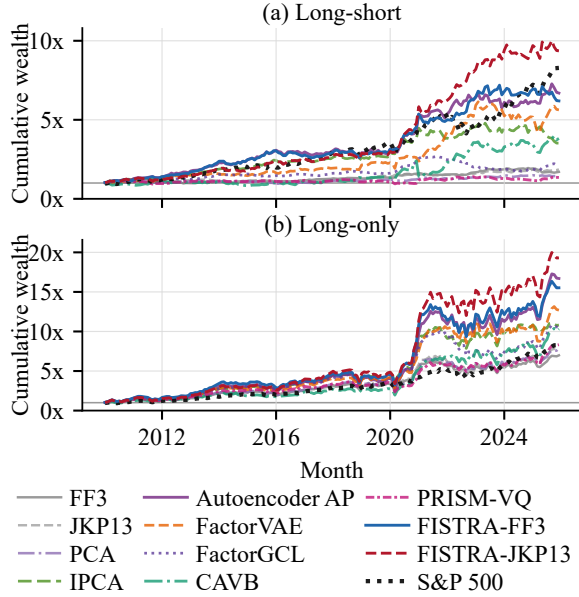


Figure 3: Cumulative wealth of stock-selection portfolios, 2010–2025, net of 25-bp costs: (a) long–short; (b) long-only.

Ablation Study (RQ2)

Table 4 reports matched component removals under both priors and a separate no-prior variant; the supplementary appendix defines each specification. Every removal degrades performance: eliminating the Set Transformer causes the largest overall loss, confirming the value of contextual cross-stock modeling, while NoPrior reduces RankIC to 1.360%, showing that established financial factors provide important guidance for learning complementary factors.

Specification	IC (%)	ICIR	RankIC (%)	RankICIR	R^2_{pred} (%)	R^2_{total} (%)
Panel A: FISTRA-FF3						
Full model	3.324	0.491	5.894	0.554	1.030	9.887
Static queries	2.441	0.354	3.589	0.370	0.596	8.048
No Sinkhorn	2.780	0.426	3.827	0.395	0.813	9.027
No set interaction	2.923	0.373	2.141	0.298	0.413	7.134
No Set Transformer	1.661	0.201	2.913	0.216	0.129	5.052
Panel B: FISTRA-JKP13						
Full model	3.709	0.543	5.295	0.511	1.340	10.406
Static queries	2.592	0.340	3.565	0.346	0.665	8.753
No Sinkhorn	3.347	0.441	3.939	0.396	0.785	8.633
No set interaction	2.930	0.309	2.596	0.288	0.342	6.663
No Set Transformer	1.495	0.362	2.188	0.219	0.215	4.656
Panel C: Observed-Factor Prior Ablation						
FISTRA-NoPrior	1.895	0.241	1.360	0.151	0.111	9.029

Table 4: Component Ablation Results

Incremental Information Beyond Established Factors (RQ3)

Because annual re-estimation leaves latent coordinates basis-indeterminate, our conclusions concern the learned span; the

basis-invariant joint test and strictly pre-test alignment are detailed in the appendix. We test whether the learned factors contain average-return information beyond their matched established factor set using $f_{k,t} = \alpha_k + \gamma_k^\top g_t + e_{k,t}$, where $f_{k,t}$ is the realized return of learned factor k , $g_t \in \mathbb{R}^{K^o}$ contains the matched FF3 or JKP13 returns, γ_k contains their exposures, and $e_{k,t}$ is the residual. The intercept α_k is the learned factor’s average return unexplained by those exposures. For $\alpha = (\alpha_1, \dots, \alpha_{K^h})^\top$, we test $H_0^{\text{joint}} : \alpha = \mathbf{0}$ against $H_1^{\text{joint}} : \alpha_k \neq 0$ for at least one k . The null means that the established factors explain every learned factor’s average return; the alternative means that at least one contains an average-return component beyond their span. We use a six-lag Newey–West Wald test; the supplementary appendix gives alignment and individual-test details.

Matched test	Wald statistic	df	p-value	Decision
FISTRA-FF3 $\rightarrow$ FF3	61.421	4	$p < 0.0001$	Reject
FISTRA-JKP13 $\rightarrow$ JKP13	52.810	4	$p < 0.0001$	Reject

Table 5: Matched Joint Spanning Tests

Both joint tests reject H_0^{joint} ($p < 0.0001$), showing that under each prior, FISTRA learns average-return information beyond the matched established-factor span.

Robustness Analysis (RQ4)

We vary the number of learned factors, the Conditional Beta Network depth, the induced set attention block (ISAB) depth, and transaction costs while holding the protocol fixed. One- and two-factor variants are weaker, whereas FISTRA-JKP13 maintains a RankIC of 5.230–5.302% with four to six factors. A third beta residual block or ISAB layer improves some IC and R^2 measures but does not consistently dominate across priors, supporting the two-block, two-ISAB setting as a stable common choice. At 50 bps, FISTRA-JKP13 retains the best long–short AR and SR (11.59%, 0.90). These results demonstrate robustness to reasonable capacity and transaction-cost choices; complete results appear in the supplementary appendix.

Conclusion

FISTRA combines established factors with a temporal Set Transformer to learn tradable complementary factors. Exposure-neutral allocation keeps them auditable while modeling nonlinear cross-stock relations. Supported by ablations, FISTRA outperforms factor and prior-informed baselines in both settings on prediction and net-of-cost investment performance; learned factors more than double annualized returns, improve Sharpe ratios, and add information beyond established spans. Minor limitations arise from omitted market-impact, capacity, and short-sale constraints and cross-vintage rotations; explicit constraints and alignment address both. In the future, we plan to incorporate these constraints and extend FISTRA to international and higher-frequency equities. Overall, FISTRA offers a strong balance of predictive flexibility, tradability, and interpretability.

References

- Andreou, E.; Gagliardini, P.; Ghysels, E.; and Rubin, M. 2025. Spanning Latent and Observable Factors. *Journal of Econometrics*, 248: 105743.
- Bai, J.; and Ng, S. 2002. Determining the Number of Factors in Approximate Factor Models. *Econometrica*, 70(1): 191–221.
- Bai, S.; Kolter, J. Z.; and Koltun, V. 2018. An Empirical Evaluation of Generic Convolutional and Recurrent Networks for Sequence Modeling. *arXiv preprint arXiv:1803.01271*.
- Cuturi, M. 2013. Sinkhorn Distances: Lightspeed Computation of Optimal Transport. In *Advances in Neural Information Processing Systems*, volume 26.
- Duan, Y.; Wang, L.; Zhang, Q.; and Li, J. 2022. FactorVAE: A Probabilistic Dynamic Factor Model Based on Variational Autoencoder for Predicting Cross-Sectional Stock Returns. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 36, 4468–4476.
- Duan, Y.; Wang, W.; and Li, J. 2025. FactorGCL: A Hypergraph-Based Factor Model with Temporal Residual Contrastive Learning for Stock Returns Prediction. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 39, 173–181.
- Fama, E. F.; and French, K. R. 1993. Common Risk Factors in the Returns on Stocks and Bonds. *Journal of Financial Economics*, 33(1): 3–56.
- Fama, E. F.; and French, K. R. 2015. A Five-Factor Asset Pricing Model. *Journal of Financial Economics*, 116(1): 1–22.
- Feng, G.; He, J.; Polson, N. G.; and Xu, J. 2024. Deep Learning in Characteristics-Sorted Factor Models. *Journal of Financial and Quantitative Analysis*, 59(7): 3001–3036.
- Giglio, S.; and Xiu, D. 2021. Asset Pricing with Omitted Factors. *Journal of Political Economy*, 129(7): 1947–1990.
- Gu, S.; Kelly, B.; and Xiu, D. 2020. Empirical Asset Pricing via Machine Learning. *Review of Financial Studies*, 33(5): 2223–2273.
- Gu, S.; Kelly, B.; and Xiu, D. 2021. Autoencoder Asset Pricing Models. *Journal of Econometrics*, 222(1): 429–450.
- He, K.; Zhang, X.; Ren, S.; and Sun, J. 2016. Deep Residual Learning for Image Recognition. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, 770–778.
- Hou, K.; Xue, C.; and Zhang, L. 2015. Digesting Anomalies: An Investment Approach. *Review of Financial Studies*, 28(3): 650–705.
- Jensen, T. I.; Kelly, B. T.; and Pedersen, L. H. 2023. Is There a Replication Crisis in Finance? *The Journal of Finance*, 78(5): 2465–2518.
- Jia, Y.; Li, G.; Zhao, G.; Lin, X.; and Li, G. 2024. GraphVAE: Unveiling Dynamic Stock Relationships with Variational Autoencoder-Based Factor Modeling. In *Proceedings of the 33rd ACM International Conference on Information and Knowledge Management*, 3807–3811.
- Kelly, B. T.; Pruitt, S.; and Su, Y. 2019. Characteristics Are Covariances: A Unified Model of Risk and Return. *Journal of Financial Economics*, 134(3): 501–524.
- Kim, N.; Ock, S. E.; and Song, J. W. 2025. FactorVQ-VAE: Discrete Latent Factor Model via Vector Quantized Variational Autoencoder. *Knowledge-Based Systems*, 318: 113460.
- Kim, N.; and Song, J. W. 2026. Vector-Quantized Discrete Latent Factors Meet Financial Priors: Dynamic Cross-Sectional Stock Ranking Prediction for Portfolio Construction. *arXiv preprint arXiv:2605.13407*. Accepted at IJCAI-ECAI 2026.
- Ledoit, O.; and Wolf, M. 2004. Honey, I Shrunk the Sample Covariance Matrix. *The Journal of Portfolio Management*, 30(4): 110–119.
- Lee, J.; Lee, Y.; Kim, J.; Kosiorek, A.; Choi, S.; and Teh, Y. W. 2019. Set Transformer: A Framework for Attention-Based Permutation-Invariant Neural Networks. In *Proceedings of the 36th International Conference on Machine Learning*, 3744–3753.
- Li, T.; Liu, Z.; Shen, Y.; Wang, X.; Chen, H.; and Huang, S. 2024. MASTER: Market-Guided Stock Transformer for Stock Price Forecasting. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 38, 162–170.
- Liu, R.; Huang, H.; and Ruf, J. 2025. Asset Pricing with Contrastive Adversarial Variational Bayes. In *Proceedings of the Thirty-Fourth International Joint Conference on Artificial Intelligence*, 7572–7579.
- Markowitz, H. 1952. Portfolio Selection. *The Journal of Finance*, 7(1): 77–91.
- Ross, S. A. 1976. The Arbitrage Theory of Capital Asset Pricing. *Journal of Economic Theory*, 13(3): 341–360.
- Uddin, A. 2025. Neural Instrumented Factorization: Learning Dynamic Asset Pricing Factors and Loadings through Characteristics Control. In *Proceedings of the 34th ACM International Conference on Information and Knowledge Management*, 2915–2924.
- Wei, Z.; Rao, A.; Dai, B.; and Lin, D. 2023. HireVAE: An Online and Adaptive Factor Model Based on Hierarchical and Regime-Switch VAE. In *Proceedings of the Thirty-Second International Joint Conference on Artificial Intelligence*, 4903–4911.
- Xia, H.; Ao, H.; Li, L.; Liu, Y.; Liu, S.; Ye, G.; and Chai, H. 2024. CI-STHPAN: Pre-Trained Attention Network for Stock Selection with Channel-Independent Spatio-Temporal Hypergraph. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 38, 9187–9195.
- Yoo, J.; Soun, Y.; Park, Y.-c.; and Kang, U. 2021. Accurate Multivariate Stock Movement Prediction via Data-Axis Transformer with Multi-Level Contexts. In *Proceedings of the 27th ACM SIGKDD Conference on Knowledge Discovery and Data Mining*, 2037–2045.
- Zaheer, M.; Kottur, S.; Ravanbakhsh, S.; Póczos, B.; Salakhutdinov, R.; and Smola, A. J. 2017. Deep Sets. In *Advances in Neural Information Processing Systems*, volume 30.

Zhao, Y.; Zhang, W.; Yang, T.; Jiang, Y.; Huang, F.; and Lim, W. Y. B. 2026. STORM: A Spatio-Temporal Factor Model Based on Dual Vector Quantized Variational Autoencoders for Financial Trading. In *Proceedings of the Nineteenth ACM International Conference on Web Search and Data Mining*, 1017–1026.